

Lab Assignment-5

Computer Architecture (CS F342)

Semester-I, 2026-27

Department of Computer Science and Information Systems (CSIS)

BITS-Pilani, K K Birla Goa Campus, Goa, India.

Due date: Sept 24, 2026: 13.00

of problems: 2

Marks: 20 (8+12)

Download the *CA_Lab-5_Template.zip* and it contains 2 problems (P1 and P2 folders) and rename it as *<CampusID>_CA_Lab-5*.

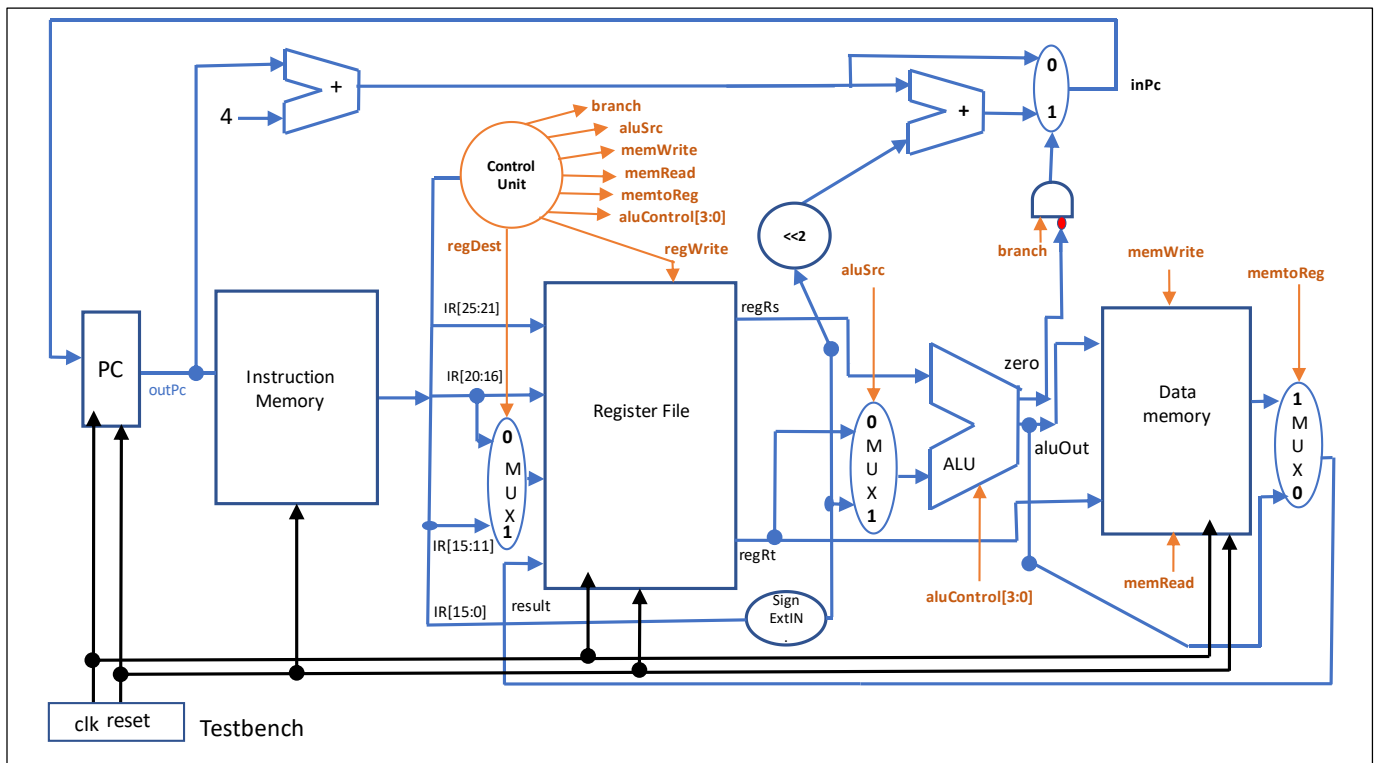
One's lab marks depend on one's performance on lab viva even if one has no complete working code.

Serious action will be taken if one has the fully working code but is unable to explain it during the viva.

P1) Consider the 32-bit MIPS instruction model (instruction format is given below) and, using Verilog HDL language, simulate the MIPS's instructions: LW, SW, ADD, ADDI, and BNE. For this, one has to define the following modules. There is no need to define other variables and modify the other module except the below in the program.

File Name	Module	Marks
ALU.v	ALU	1
ALU.v	aluControl	1
instrMem.v	Dff_IM	0.3
instrMem.v	register_IM	0.2
controller.v	controlCircuit	2
topModule.v	singleCycle	3
testbench.v	testbench	0.5

MIPS's single-cycle model

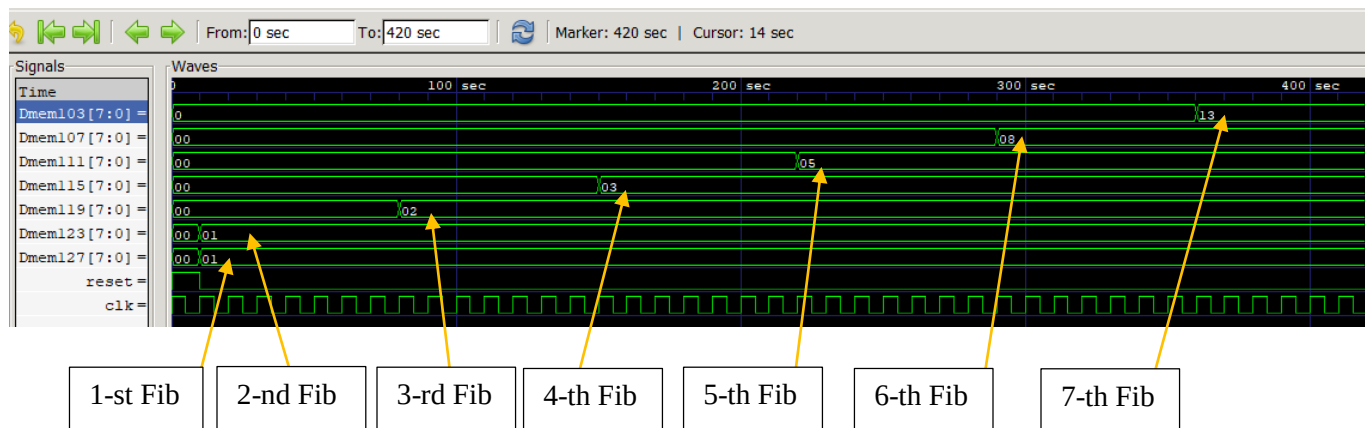


MIPS has 32 registers and R0 holds zero value only.

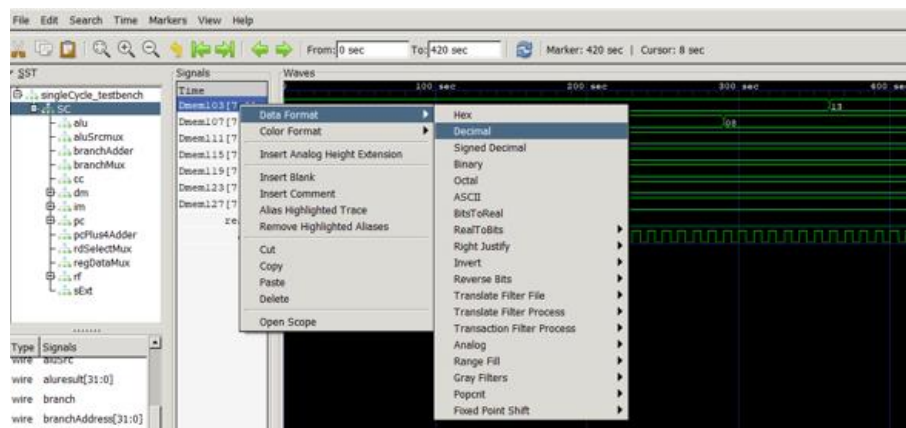
Instruction format:

R-Type	OPCODE						RS					RT					RD					SHAMT					FUNCT						
	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
I-Type	OPCODE						RS					RT/RD					IMMEDIATE																
	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	

A test program, stored in instruction memory, generates 7 Fibonacci numbers in the template. The program is written using the above-mentioned instructions. The first two Fibonacci numbers are stored in data memory at locations 124 and 120. The program calculates the first seven Fibonacci numbers, and the remaining five Fibonacci numbers can be stored in locations 116 onwards. The Big-endian format is considered to store the instruction and data. Memory is byte-addressable. Separate memory is used for the instructions and data. The memory contains 128 locations. However, the program counter (PC) is 32-bit. Some bits of the PC are used to identify the memory locations. Similarly, for data memory. The program generates the following outputs:



Following is changing procedure for signals' data format as decimal:



P2) Write an ASM code using MIPS ISA to find out the first 7 Tribonacci numbers and store the program in the instruction memory and data into the data memory as in problem P1. Display the Tribonacci numbers in the GTKWave. Upload the output and Verilog codes in the corresponding folder. **(Marks: 3 + 8 + 1)**

Instructions for uploading the results:

Download the *CA_Lab-5_Template.zip* and rename the folder as *<CampusID>_CA_Lab-5*. Output must be stored as *<CampusID>_output.png* in the P1 and P2 folder. Create a zip file *<CampusID>_CA_Lab-5.zip* which contains all the file mention above and submit it in Quanta.

Do not create archives in other formats (**rar, tar.gz etc**). Once uploaded on Quanta, remember to **submit for grading**. **Do not** leave it as a draft.

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Note: Please don't upload the assignments, template file/solution and lab. manual on GitHub or others public repository.

It violates the BITS's Intellectual Property Rights (IPR).

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